

JAEMIN SONG

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EDUCATION

Texas A&M University

M.S. in industrial engineering

Jan. 2025 – Dec. 2026(Expected), College Station, TX

Hongik University

Seoul, Korea

B.S. in Industrial and Data Engineering (GPA: 3.68/4.5)

Mar. 2017 – Feb. 2024

- Graduation Project: "AI-based Diagnosis of Laryngeal Cancer using Voice Data for Enhanced Diagnosis and Treatment"
- Coursework: Data Programming and Practices, Application of Data Analysis, Data Science, Spatial Big Data Analysis, Design of Human Sensibility Interface, Fundamentals of Internet of Things, Introduction to Data mining, Machine Learning and Smart City, Database, System Analysis, Visual Perception and Image
- Academic Scholarship for 4 semesters (Fall 2017, Fall 2018, Spring 2019, Fall 2019)

WORK EXPERIENCE

ELORILAB

Seoul, Korea

Internship

2024. Sep – 2024. Dec

- Interned at a startup specializing in hyperspectral technology for contaminant classification in food industry.
- Responsibilities included collecting and preprocessing data to develop solutions for specific food products, performing calibration for model application, and reporting on-site challenges to researchers for further analysis.

RESEARCH EXPERIENCE

Hongik University Artificial Intelligence Application (AIA) Lab

Seoul, Korea

Research Assistant (PI: Dr. Yong Oh Lee)

Feb. 2024 – Aug. 2024

"Improving Laryngeal Cancer Detection: Comparing Octave-band Filtering & MFCC in Voice Data Machine Learning"

- Applied Octave Frequency Cepstrum (OFC) using an Octave Band Filter in place of the previously utilized Mel-Frequency Cepstral Coefficients (MFCCs).

Undergraduate Research Assistant (PI: Dr. Yong Oh Lee)

Dec. 2022 – Feb. 2024

"AI-based Diagnosis of Laryngeal Cancer using Voice Data for Enhanced Diagnosis and Treatment"

- Digitized analog voice frequencies by applying signal processing techniques such as Fourier transform, short-term Fourier transform, and discrete cosine transform to preprocess voice data.
- Calculated Mel-frequency Cepstral Coefficients (MFCCs) from vowel sound voice samples and converted the feature-engineered data into images for a proposed computer vision model, developing a novel feature engineering method of human voice data collected from laryngeal cancer patients by.
- Trained ResNet50 encoder via various ML classification models such as Extreme Gradient Boosting (XGBoost), Support Vector Machine, Artificial Neural Network, and Residual Neural Network for hypothesis testing and ablation studies.
- Extracted acoustic features such as jitter, shimmer, and HNR from voice data using Praat to overcome MFCC limitations.

"Predicting Chemical Properties using BiLSTM with BPE tokenization method using SMILES Representation"

- Developed a data-driven model using molecular data to predict compound properties for pharmaceutical companies to accelerate drug discovery and development.
- Tokenized Simplified Molecular Input Line Entry System (SMILES) data by reinventing a traditional data compression algorithm, "Byte Pair Encoding" (BPE) for Natural Language Processing (NLP) tokenization method.
- Improved the BPE algorithm for tokenizing SMILES data to use LSTM architecture and successfully tokenized data.
- Trained a Bidirectional LSTM (BiLSTM) model as a baseline of the proposed sequence classification model to classify toxicity with tokenized molecular data (tox21) using my tokenization pipeline.

JOURNAL PUBLICATIONS

J. Song, H. Kim, and Y.O. Lee. "Laryngeal Disease Classification Using Voice Data: Octave-Band vs. Mel-Frequency Filters." Heliyon, 2024.

H.B. Kim, J. Song, S.H. Park, and Y.O. Lee. "Classification of Laryngeal Diseases including Laryngeal Cancer, Benign Mucosal Disease, and Vocal Cord Paralysis by Artificial Intelligence using Voice Analysis." *Scientific Reports*, 2023.

PRESENTATIONS

Conference

J.M. Song, et al. "Enhancing Vocal-Based Laryngeal Cancer Screening with Additional Patient Information and Voice Signal Embedding." Big Data Analytics for Health and Medicine (BDA4HM 2023), Workshop at *2023 IEEE International Conference on Big Data (IEEE Big Data 2023)*, 2023.

Oral Presentation

"Prediction and Analysis Utilization Methods of Seoul's Popular Places," Korea Data Agency (KDATA), Seoul, Korea, 2022.

HONORS & AWARDS

2nd place, Bio-health Data Competition, National Research Foundation of Korea 2023

- Developed a predictive model using Convolutional Neural Network (CNN) techniques on dental X-ray images to assess the risk of alveolar nerve damage post wisdom tooth extraction.

1st place, Bio-health Academic Competition, National Research Foundation of Korea 2023

- Recognized for the project "Voice-based Deep Learning Model for Screening Laryngeal Diseases," demonstrating an innovative application of deep learning in medical diagnostics.

3rd place, Data Science Competition, Korean Data Agency affiliated by Government of Korea 2023

- Developed a machine learning model utilizing SVM, LightGBM, XGBoost, Decision Tree, and KNN to analyze the traits of existing popular places in Seoul city, forecasting future popular tourist locations.

TECHNICAL SKILLS

Programming Language: Python, R, C, C++,

Frameworks and Packages: TensorFlow, PyTorch, Scikit-Learn, NumPy, Librosa

Computer and Data Engineering: SQL

EXTRACURRICULAR ACTIVITIES

Hongik University (HIU) **Seoul, Korea**

Member, Entrepreneurship Club 2018-2019

- Participated in the "Hongik University Startup Competition" May 2019
- Participant in the "H-biz Hackathon" Entrepreneurship Camp Sep. 2018
- Attended the "Hongik University Entrepreneurship Festival" May 2018

Member, Student Council 2017-2018

Member, Astronomical Observation Club 2017-2019